University of the Philippines Diliman
College of Engineering
Electrical and Electronics Engineering Institute

Capacitors and Inductors

1st Semester, A.Y. 2026–2027

Student Outcomes

- Recall the properties and behavior of capacitors and inductors, and their voltage-current relationship
- Calculate the voltage, current, power, and energy of capacitors and inductors
- Simplify a network of capacitors and inductors

Lecture Outline

- Fundamental Characteristics and Operation of Capacitors
- Fundamental Characteristics and Operation of Inductors
- Important characteristics of basic elements

Capacitors

A **capacitor** consists of two conducting plates separated by an insulator or dielectric. It stores energy in its electric field.

Common applications:

- Tuning circuits of radio receivers
- Dynamic memory elements

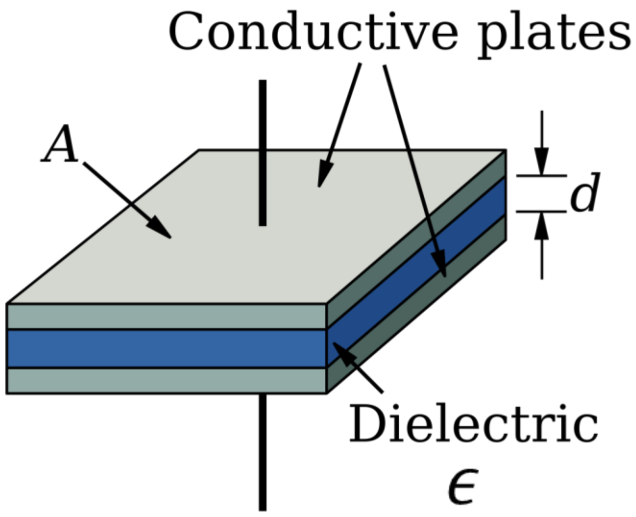
Figure. Different capacitors. Image reproduced from Wikimedia.

Capacitance

ratio of the **charge on one plate** of a capacitor to the **voltage between the two plates**, measured in farads (F)

$$C = \frac{q}{v}$$

capacitance depends on its **physical dimensions!**



$$C = \epsilon \frac{A}{d}$$

Figure. Parallel-plate capacitor. Image reproduced from Wikimedia.

Current-voltage relationship of capacitors

capacitance

$$C = \frac{q}{v}$$

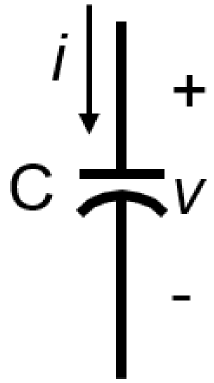
current

$$i = \frac{dq}{dt}$$

What is the current-voltage relationship in terms of voltage?

current-voltage relationship

$$i = C\frac{dv}{dt}$$



Power and energy of capacitors

- The power **delivered** to the capacitor is defined by the voltage-current product:

$$\mathbf{p(t)} = \mathbf{v_C(t) i_C(t)} = \mathbf{v_C(t)} \left(\mathbf{C \frac{dv_C(t)}{dt}} \right)$$

- The energy **stored** in the capacitor is given by:

$$\int_{t_0}^{t} p(t)\, dt = \int_{t_0}^{t} v_C(t) \left(C \frac{dv_C(t)}{dt} \right) dt = C \int_{t_0}^{t} v_C(t) dv_C(t)$$

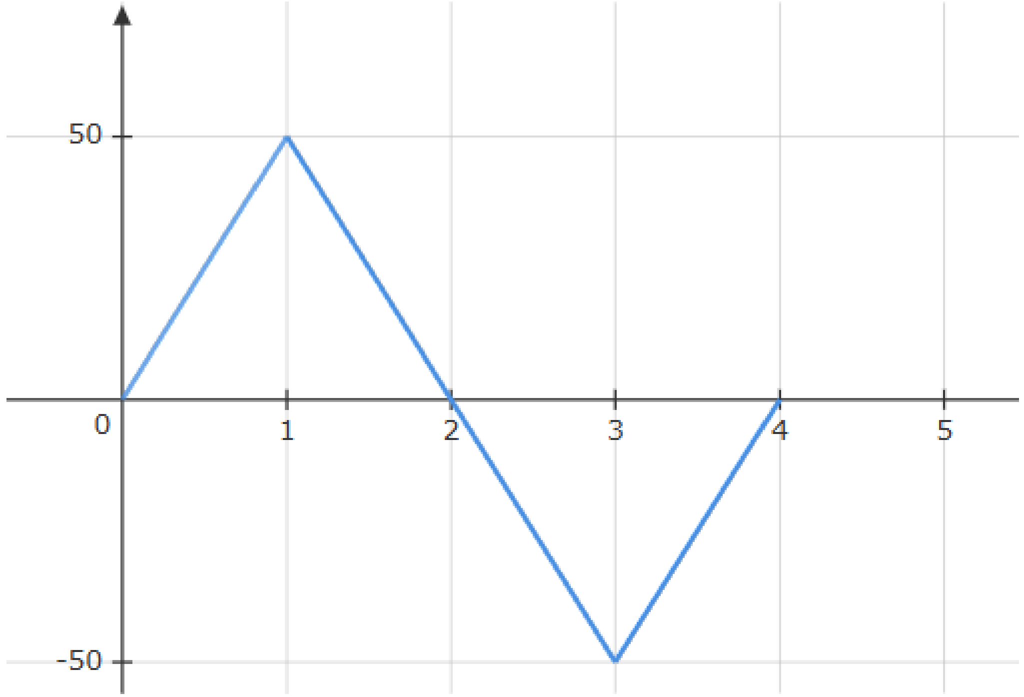
$$w_C(t) - w_C(t_0) = \frac{1}{2} C \left([v_C(t)]^2 - [v_C(t_0)]^2 \right)$$

Remarks on the properties of capacitor

- When the **voltage across a capacitor is not changing with time**, the **current through the capacitor is zero**.
- The voltage on the capacitor **must be continuous.**
- The **ideal capacitor does not dissipate energy.**
- A real, non-ideal capacitor has a **parallel-mode leakage resistance.**

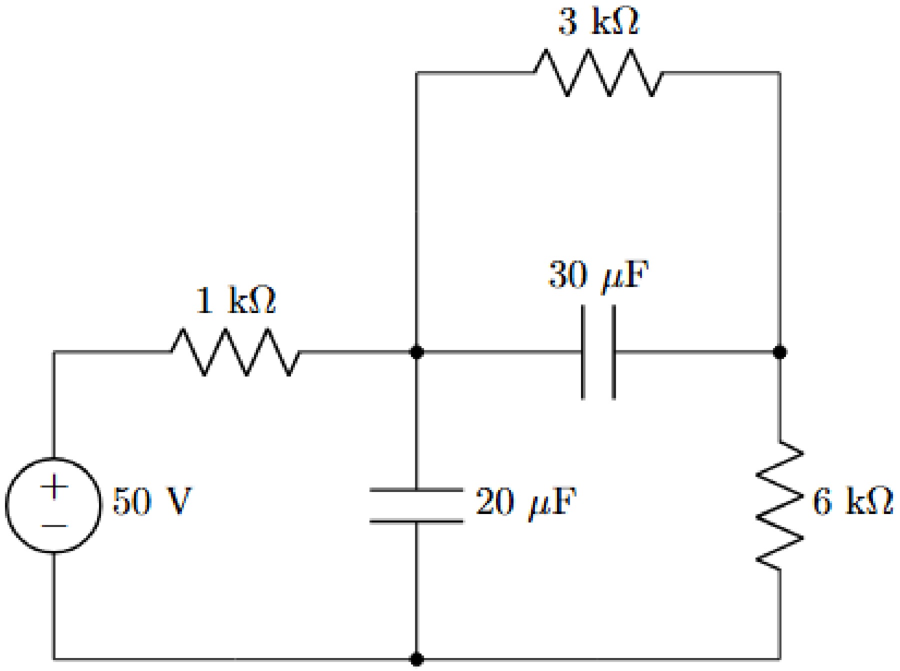
Example: Voltage-Current Curve

Determine the current through a 200 μF capacitor whose voltage is shown in the graph below.



Example: Capacitors in DC Circuits

Obtain the energy stored in each capacitor shown in the figure under DC conditions.



Parallel Capacitors

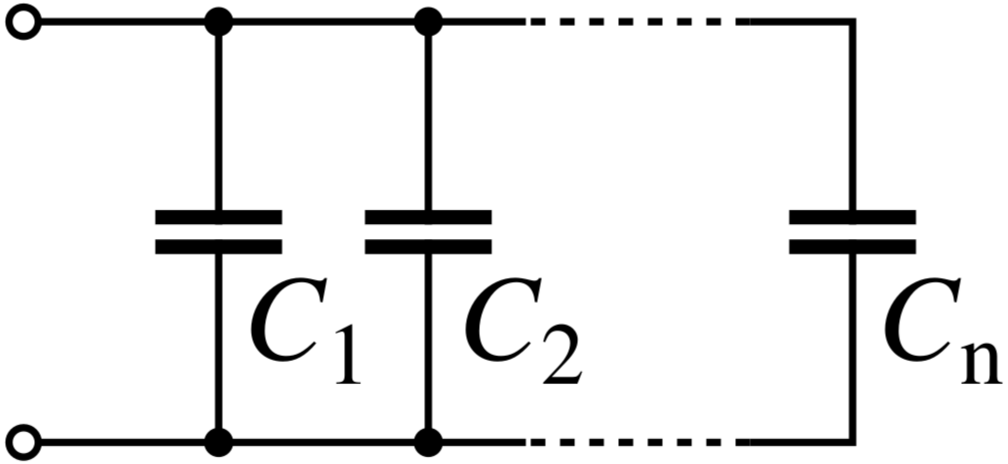


Figure. Parallel capacitors. Image reproduced from Wikimedia.

$$\mathbf{i} = \mathbf{i}_1 + \mathbf{i}_2 + \mathbf{i}_3 + \cdots + \mathbf{i}_N$$

$$\mathbf{i} = C_1 \frac{dv}{dt} + C_2 \frac{dv}{dt} + C_3 \frac{dv}{dt} + \cdots + C_N \frac{dv}{dt}$$

$$\mathbf{i} = \frac{dv}{dt} \sum_{i=1}^{N} C_i$$

$$C_{eq} = \sum_{i=1}^{N} C_i$$

Series Capacitors

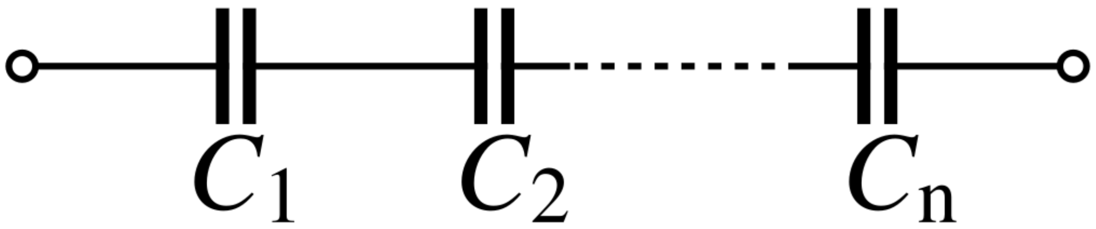


Figure. Series capacitors. Image reproduced from Wikimedia.

$$\mathbf{v} = \mathbf{v}_{C1} + \mathbf{v}_{C2} + \mathbf{v}_{C3} + \cdots + \mathbf{v}_{CN}$$

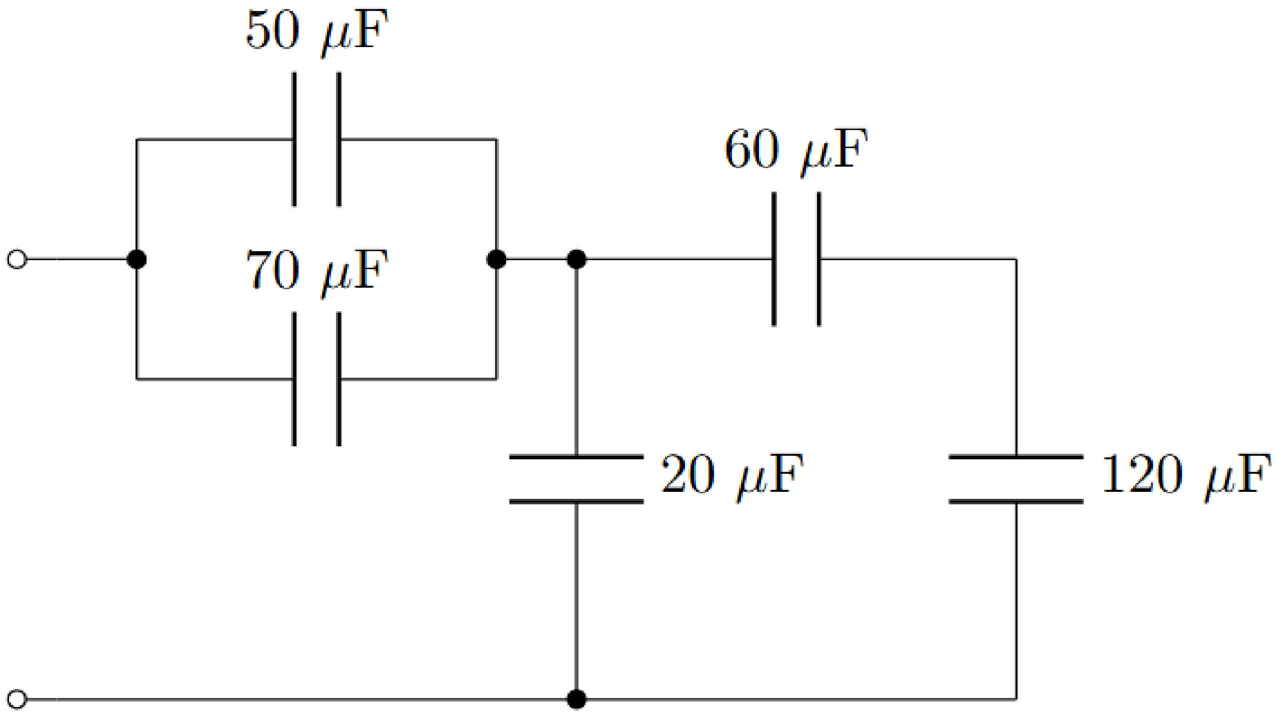
$$\mathbf{v} = \frac{1}{C1}\int_{t_0}^{t} \mathbf{i}(\tau)d\tau + \cdots + \frac{1}{CN}\int_{t_0}^{t} \mathbf{i}(\tau)d\tau$$

$$\mathbf{i} = \sum_{i=1}^{N} \frac{1}{C_i}\left(\int_{t_0}^{t} \mathbf{i}(\tau)d\tau\right)$$

$$C_{eq} = \left(\sum_{i=1}^{N} \frac{1}{C_i}\right)^{-1}$$

Example: Equivalent Capacitance

Find the equivalent capacitance seen at the terminals of the circuit shown.



Inductors

An **inductor** consists of a coil of conducting wire. It stores energy in its magnetic field.

Common applications:

- Electronic and power systems
- Communication systems

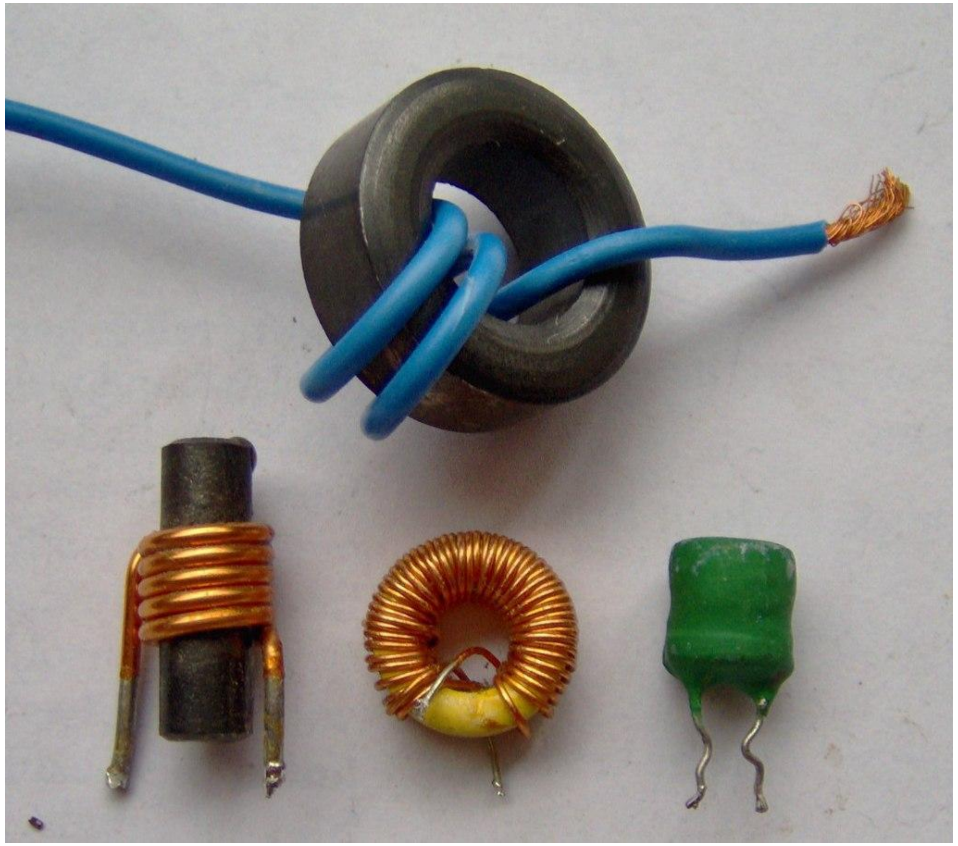


Figure. Different inductors. Image reproduced from Wikimedia.

Inductance

ratio of the **magnetic flux linkage** generated by a given **current flowing through the inductor**, measured in henrys (H)

$$L = \frac{\phi_B}{i}$$

inductance depends on its **physical dimensions!**

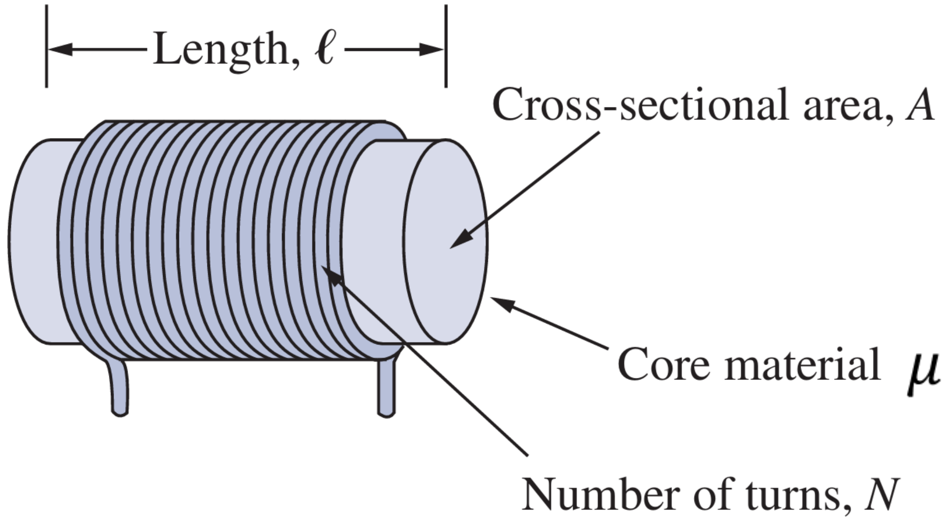


Figure. Typical form of an inductor. Image reproduced from electrical-engineering-portal.

$$L = \frac{N^2 \mu A}{\ell}$$

Current-voltage relationship of inductors

Faraday's law of induction

$$v = \frac{d\phi_B}{dt}$$

modified to accommodate passive sign convention

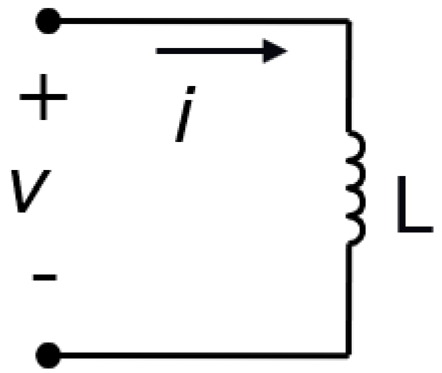
inductance

$$L = \frac{\phi_B}{i}$$

What is the current-voltage relationship in terms of current?

current-voltage relationship

$$v = L\frac{di}{dt}$$



Power and energy of inductors

- The power **delivered** to the inductor is defined by the voltage-current product:

$$\mathbf{p(t)} = v_L(t) i_L(t) = \left(L \frac{di_L(t)}{dt} \right) i_L(t)$$

- The energy **stored** in the inductor is given by:

$$\int_{t_0}^{t} p(t)\, dt = \int_{t_0}^{t} \left(L \frac{di_L(t)}{dt} \right) i_L(t) dt = L \int_{t_0}^{t} i_L(t) di_L(t)$$

$$w_L(t) - w_L(t_0) = \frac{1}{2} L \left([i_L(t)]^2 - [i_L(t_0)]^2 \right)$$

Remarks on the properties of inductor

- When the **voltage across an inductor is zero**, then **current through the inductor is constant**.

- The current through an inductor **must be continuous.**

- The **ideal inductor does not dissipate energy.**

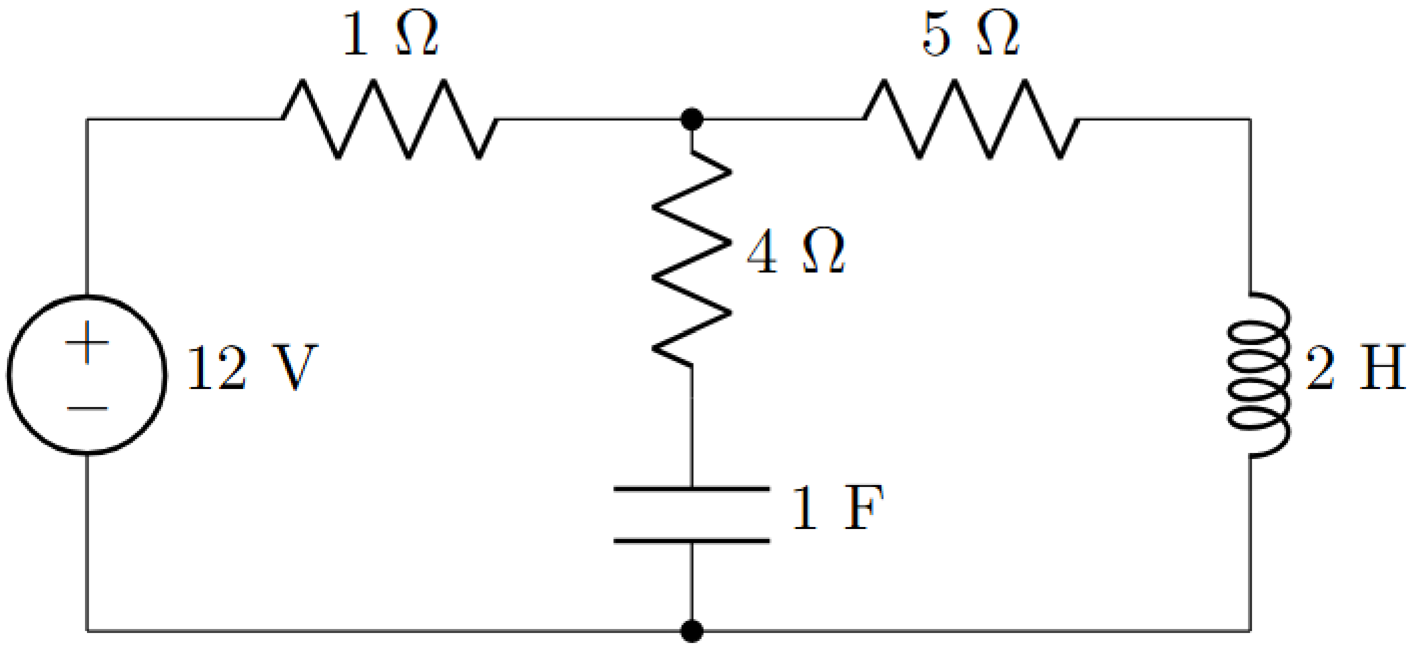
- A real, non-ideal inductor has a **winding resistance.**

Example: Voltage Equation

The terminal voltage of a 2 H inductor is $v_L(t) = 10(1 - t)$ V. Find the current flowing through it at t=4 s and the energy stored in it at t=4 s. Assume $i_L(0) = 2$ A.

Example: Inductors (and Capacitors) in DC Circuits

Obtain the energy stored in the capacitor and inductor shown in the figure under DC conditions.



Parallel Inductors

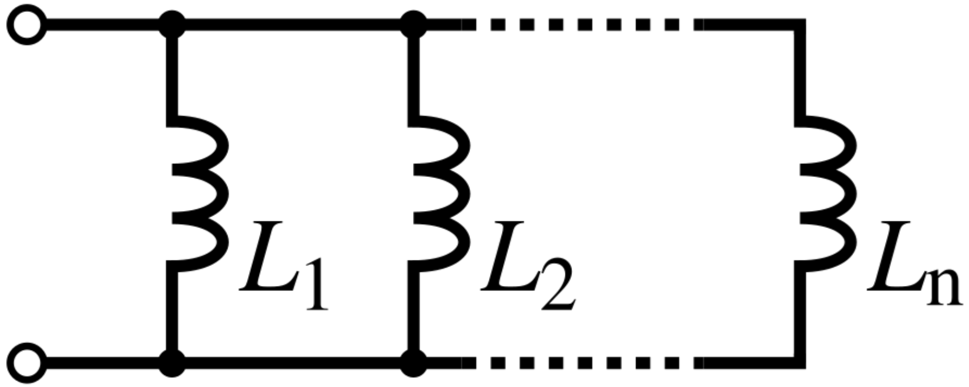


Figure. Parallel inductors. Image reproduced from Wikimedia.

$$\mathbf{i} = \mathbf{i}_1 + \mathbf{i}_2 + \mathbf{i}_3 + \cdots + \mathbf{i}_N$$

$$\mathbf{i} = \frac{1}{L1}\int_{t_0}^{t} \mathbf{v}(\boldsymbol{\tau})d\boldsymbol{\tau} + \cdots + \frac{1}{LN}\int_{t_0}^{t} \mathbf{v}(\boldsymbol{\tau})d\boldsymbol{\tau}$$

$$\mathbf{i} = \sum_{i=1}^{N} \frac{1}{\mathrm{L}_i}\left(\int_{t_0}^{t} \mathbf{v}(\boldsymbol{\tau})d\boldsymbol{\tau}\right)$$

$$L_{eq} = \left(\sum_{i=1}^{N} \frac{1}{\mathrm{L}_i}\right)^{-1}$$

Series Inductors

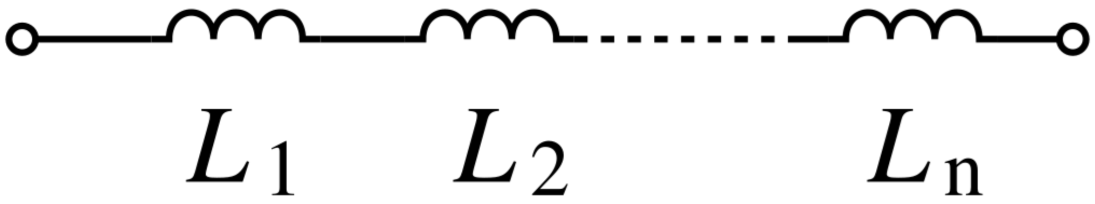


Figure. Series inductors. Image reproduced from Wikimedia.

$$\mathbf{v} = \mathbf{v}_{L1} + \mathbf{v}_{L2} + \mathbf{v}_{L3} + \cdots + \mathbf{v}_{LN}$$

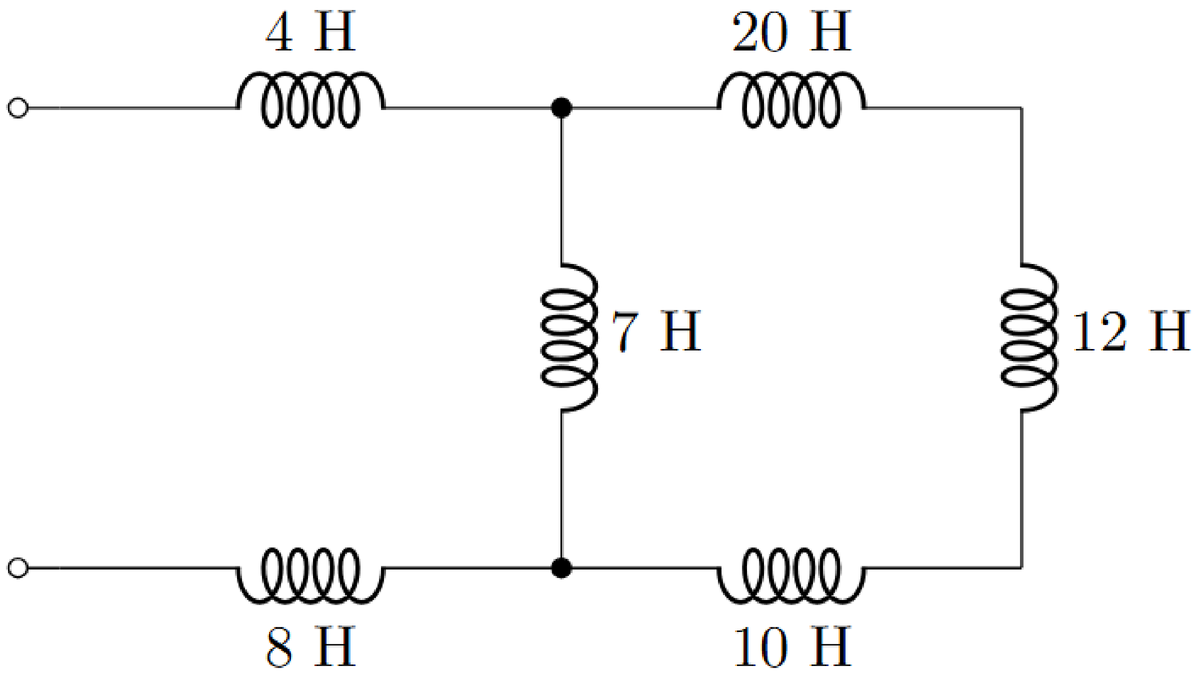
$$\mathbf{v} = \mathbf{L}_1 \frac{\mathbf{di}}{\mathbf{dt}} + \mathbf{L}_2 \frac{\mathbf{di}}{\mathbf{dt}} + \mathbf{L}_3 \frac{\mathbf{di}}{\mathbf{dt}} + \cdots + \mathbf{L}_N \frac{\mathbf{di}}{\mathbf{dt}}$$

$$\mathbf{v} = \frac{\mathbf{di}}{\mathbf{dt}} \sum_{i=1}^{N} \mathbf{L}_i$$

$$L_{eq} = \sum_{i=1}^{N} \mathbf{L}_i$$

Example: Equivalent Inductance

- Find the equivalent inductance seen at the terminals of the circuit shown.



Important characteristics of the basic elements

Relation	Resistor (R)	Capacitor (C)	Inductor (L)
v-i:	$v = iR$	$v = \frac{1}{C}\int_{t_0}^{t} i(\tau)d\tau + v(t_0)$	$v = L\frac{di}{dt}$
i-v:	$i = v/R$	$i = C\frac{dv}{dt}$	$i = \frac{1}{L}\int_{t_0}^{t} v(\tau)d\tau + i(t_0)$
p or w:	$p = i^2R = \frac{v^2}{R}$	$w = \frac{1}{2}Cv^2$	$w = \frac{1}{2}Li^2$
Series:	$R_{eq} = R_1 + R_2$	$C_{eq} = \frac{C_1C_2}{C_1 + C_2}$	$L_{eq} = L_1 + L_2$
Parallel:	$R_{eq} = \frac{R_1R_2}{R_1 + R_2}$	$C_{eq} = C_1 + C_2$	$L_{eq} = \frac{L_1L_2}{L_1 + L_2}$
At dc:	Same	Open circuit	Short circuit
Circuit variable that cannot change abruptly:	Not applicable	v	i

Summary

- Capacitors resist changes in voltage and store energy in their electric field.
- Inductors resist changes in current and store energy in their magnetic field.
- Under steady-state DC conditions, a capacitor acts as an open circuit while an inductor acts as a short circuit.
- However, an unenergized capacitor acts as a short circuit while an unenergized inductor acts as an open circuit.